

Emergent Causal Order from Entanglement

A Pedagogical Introduction to a Tensor-Network Hypothesis Test

Abstract

A common slogan in modern theoretical physics is that “spacetime is built from entanglement”. Stripped of mystique, the claim is that the causal and geometric structure we see in the macroscopic world might be a derived feature of how a quantum state is correlated across its parts. This paper presents, from first principles, one concrete way of taking that slogan seriously enough to test on a small computer. We work in 1+1 dimensional lattice quantum electrodynamics, where a tensor-network simulation can produce the full quantum state of an evolving particle–antiparticle pair on a 20-site chain. From that state we read off three different partial orders on space-time events: the order induced by the lattice’s own causal structure, the order induced by the Lieb–Robinson bound, and an order built from the entanglement structure via Nielsen majorization. The hypothesis we test is whether the three orders coincide. The paper builds every ingredient from undergraduate linear algebra and quantum mechanics; no outside references are required to follow the argument or interpret the result.

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1 Introduction

The question we want to answer is a small, sharply-posed, numerical version of a much grander question. Let us start with the grand version and then reduce it.

The grand question. General relativity says spacetime has a causal structure: every point has a past, a future, and a set of points spacelike-separated from it. Quantum mechanics says quantum states can be entangled: knowing part of a state restricts what the other part can be. Various lines of research over the past two decades have suggested that these are not two independent facts, but that the causal structure of spacetime might be *built out of* the entanglement structure of an underlying quantum state. The most ambitious version of the claim is that there is no spacetime at the bottom layer of physics; there are only quantum correlations, and what we call “causality” is a coarse-grained reading of those correlations.

Why we cannot directly test it. There is no version of quantum gravity we trust well enough to simulate. Even if there were, the gulf between Planck-scale physics and the experimental scales we can probe is enormous. Any direct test of the grand claim is, for the foreseeable future, out of reach.

The small question. We can, however, ask a much more limited question that is in principle testable today. Take any quantum many-body system on a discrete lattice. Such a system has its own intrinsic causal structure, given by the lattice on which the Hamiltonian acts. It also has, via the Lieb–Robinson theorem, a strict speed limit on how fast information can propagate. A real-time-evolved state of such a system is entangled, and the entanglement, when measured between any two contiguous regions, has a quantitative structure: the Schmidt spectrum. We will define an ordering relation among space-time “cuts” based purely on those Schmidt spectra. And we will ask the following:

Is the partial order one reads off from the entanglement structure the same as the partial order one reads off from causality?

This question is finite, well-posed, and amenable to a tensor-network simulation that runs in under a minute on a laptop. This paper is about how to set up that test, what the right answer would tell us, and what the actual answer turns out to be.

Roadmap. Sections 2–4 build the mathematical machinery: probability vectors, partial orders, density matrices, the Schmidt decomposition, and majorization. Section 5 introduces the lattice Schwinger model, the smallest non-trivial gauge theory, as the physical test bed. Section 6 explains the Lieb–Robinson bound and causal sets. Section 7 brings the three partial orders together and states the hypothesis precisely. Sections 8–9 describe the numerical machinery and comparison statistics. Section 10 presents the actual result of the experiment. Sections 11–12 interpret it and lay out what would distinguish the two natural readings. An appendix collects notation.

Prerequisites. Linear algebra over the complex numbers; the bra–ket formalism of quantum mechanics at the level of a first undergraduate course; a willingness to take a few canonical results (Lieb–Robinson, Nielsen’s majorization theorem, the existence of efficient matrix-product-state algorithms) on faith. Where a result is stated without proof we will say so explicitly and explain enough about its content that the rest of the paper does not depend on the proof.

2 Mathematical preliminaries

2.1 Probability vectors

A *probability vector* of length d is a tuple $p = (p_1, \dots, p_d)$ with $p_i \geq 0$ and $\sum_i p_i = 1$. Two probability vectors of different lengths can be compared by zero-padding the shorter to the length of the longer; we adopt this convention throughout. The set of probability vectors on d symbols is the standard $(d-1)$ -simplex Δ_{d-1} .

Two extremes are useful to keep in mind:

- the *point distribution* $(1, 0, \dots, 0)$, corresponding to a deterministic outcome;
- the *uniform distribution* $(\frac{1}{d}, \dots, \frac{1}{d})$, corresponding to maximal uncertainty among d outcomes.

We will see that pure quantum states attached to bipartitions produce probability vectors via their Schmidt spectrum.

2.2 Partial orders

A *partial order* on a set S is a relation $\preceq$ such that for all $a, b, c \in S$:

- (i) $a \preceq a$ (reflexivity);
- (ii) $a \preceq b$ and $b \preceq a$ imply $a = b$ (antisymmetry);
- (iii) $a \preceq b$ and $b \preceq c$ imply $a \preceq c$ (transitivity).

We say a and b are *comparable* if $a \preceq b$ or $b \preceq a$, and *incomparable* otherwise. The familiar $\leq$ on real numbers is a *total order* (every pair is comparable); a partial order need not be total.

A partial order on a finite set is conveniently visualized by its *Hasse diagram*: a graph with one vertex per element and an edge $a \rightarrow b$ whenever $a \preceq b$ and there is no $c \neq a, b$ with $a \preceq c \preceq b$ (we say a is *covered by* b). Transitive consequences are implied by following multiple edges.

Example 2.1 (Subsets of a 3-element set). Take $S = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}$ ordered by $\subseteq$. The Hasse diagram is shown in Figure 1. Note that $\{2\}$ and $\{1, 3\}$ are incomparable: neither contains the other.

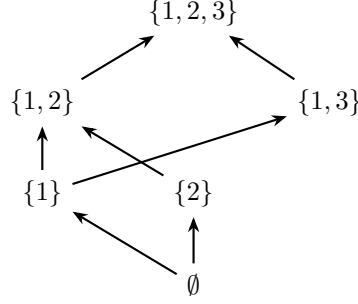


Figure 1: Hasse diagram of a partial order on six subsets, ordered by $\subseteq$. Edges show the cover relation; transitive consequences are implied. $\{2\}$ and $\{1, 3\}$ are incomparable.

2.3 Comparing two partial orders

Given two partial orders $\preceq_a$ and $\preceq_b$ on the same finite set S , we want a number that tells us how similar they are. For each unordered pair $\{x, y\} \subseteq S$ with $x \neq y$, exactly one of the following holds in each order:

- (a) x relates to y (i.e. $x \preceq y$ or $y \preceq x$);
- (b) x and y are incomparable.

For pairs comparable in *both* orders we can further ask whether they relate the same way (*concordant*) or in opposite ways (*discordant*). Define

$$n_{cc} = \#\{\text{pairs comparable in both, agreeing in direction}\}, \quad (1)$$

$$n_{dd} = \#\{\text{pairs comparable in both, disagreeing in direction}\}, \quad (2)$$

$$n_a^o = \#\{\text{pairs comparable in } \preceq_a \text{ only}\}, \quad (3)$$

$$n_b^o = \#\{\text{pairs comparable in } \preceq_b \text{ only}\}. \quad (4)$$

The classical *Kendall correlation coefficient* on the both-comparable subset is

$$\tau_{ab} = \frac{n_{cc} - n_{dd}}{n_{cc} + n_{dd}} \in [-1, 1]. \quad (5)$$

A value $\tau_{ab} = +1$ means the two orders agree wherever both relate the pair; $\tau_{ab} = -1$ means they disagree everywhere; values near 0 mean the agreement and disagreement are roughly balanced.

The numbers n_a^o and n_b^o measure asymmetric disagreement: “ $\preceq_a$ relates this pair, $\preceq_b$ does not”. We will need both τ and the asymmetric counts in Section 9; τ is informative when both orders are dense, n_a^o and n_b^o are informative when one order is much sparser than the other.

3 Quantum states and the Schmidt decomposition

3.1 Pure states

A quantum system has a complex Hilbert space $\mathcal{H}$. A pure state is a unit vector $|\psi\rangle \in \mathcal{H}$, defined up to an overall complex phase. Inner products are written $\langle\phi|\psi\rangle$. Operators $O : \mathcal{H} \rightarrow \mathcal{H}$ have an adjoint $O^\dagger$; *Hermitian* operators ($O = O^\dagger$) play the role of observables; *unitary* operators ($UU^\dagger = I$) generate dynamics.

For finite-dimensional $\mathcal{H} \cong \mathbb{C}^d$ we will routinely fix an orthonormal basis $\{|i\rangle\}_{i=1}^d$ and write $|\psi\rangle = \sum_i c_i |i\rangle$ with $\sum_i |c_i|^2 = 1$.

3.2 Composite systems

If two systems have Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, the combined system has Hilbert space $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$, the tensor product. Concretely, if $\dim \mathcal{H}_A = d_A$ and $\dim \mathcal{H}_B = d_B$, then $\dim \mathcal{H}_{AB} = d_A \cdot d_B$, and a basis is $\{|i\rangle_A \otimes |j\rangle_B\}_{i,j}$, abbreviated $|i, j\rangle$.

A general state of the joint system is

$$|\Psi\rangle = \sum_{i,j} \Psi_{ij} |i, j\rangle, \quad \sum_{i,j} |\Psi_{ij}|^2 = 1. \quad (6)$$

The matrix Ψ of coefficients carries the entire correlation structure between A and B .

A state is called a *product state* if it factorizes: $|\Psi\rangle = |\phi\rangle_A \otimes |\chi\rangle_B$, equivalently $\Psi_{ij} = \phi_i \chi_j$. Otherwise it is *entangled*.

3.3 Reduced density matrices

If we have access only to subsystem A of the joint state $|\Psi\rangle$, the relevant object is the *reduced density matrix*

$$\rho_A = \text{Tr}_B |\Psi\rangle\langle\Psi|, \quad (7)$$

where Tr_B is the partial trace over $\mathcal{H}_B$. Concretely, $(\rho_A)_{ii'} = \sum_j \Psi_{ij} \overline{\Psi_{i'j}}$, i.e. $\rho_A = \Psi \Psi^\dagger$ (treating Ψ as a $d_A \times d_B$ matrix). ρ_A is Hermitian, positive semidefinite, and has trace 1. Its eigenvalues form a probability vector.

Symmetrically, $\rho_B = \text{Tr}_A |\Psi\rangle\langle\Psi| = \Psi^\dagger \Psi$ has the same nonzero eigenvalues as ρ_A . This is a non-trivial fact whose proof is the next subsection.

3.4 The Schmidt decomposition

Apply the singular value decomposition (SVD) to the coefficient matrix Ψ :

$$\Psi = U S V^\dagger, \quad (8)$$

where U is $d_A \times r$ with orthonormal columns, V is $d_B \times r$ with orthonormal columns, $S = \text{diag}(s_1, \dots, s_r)$ with $s_1 \geq s_2 \geq \dots \geq s_r > 0$, and $r \leq \min(d_A, d_B)$ is the *Schmidt rank* of $|\Psi\rangle$.

Define new orthonormal vectors $|u_\alpha\rangle_A = \sum_i U_{i\alpha} |i\rangle_A$ and $|v_\alpha\rangle_B = \sum_j \overline{V_{j\alpha}} |j\rangle_B$. Then plugging the SVD back into (6) yields the *Schmidt decomposition*

$$|\Psi\rangle = \sum_{\alpha=1}^r s_\alpha |u_\alpha\rangle_A \otimes |v_\alpha\rangle_B, \quad \sum_{\alpha} s_\alpha^2 = 1. \quad (9)$$

The numbers $\lambda_\alpha = s_\alpha^2$ are the *Schmidt coefficients*. They are exactly the nonzero eigenvalues of both ρ_A and ρ_B :

$$\text{spec}(\rho_A) = \text{spec}(\rho_B) = (\lambda_1, \dots, \lambda_r, 0, \dots, 0). \quad (10)$$

The vector $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_r)$, sorted in nonincreasing order and zero-padded to a chosen length, is the *Schmidt spectrum* or *entanglement spectrum* of the bipartition. It is a probability vector in the sense of Section 2, and it is the central object of this paper.

Remark. A pure state is a product state iff its Schmidt rank is 1, iff its Schmidt spectrum is the point distribution $(1, 0, \dots, 0)$. A pure state is *maximally entangled* iff its Schmidt spectrum is uniform on $\min(d_A, d_B)$ symbols. Everything else is in between.

3.5 Entanglement entropy

The natural scalar summary of λ is its Shannon entropy

$$S(\rho_A) = - \sum_{\alpha} \lambda_{\alpha} \log \lambda_{\alpha}, \quad (11)$$

called the *entanglement entropy* of the bipartition. It ranges from 0 (product state) to $\log r$ (maximally entangled). Two different states can have the same entanglement entropy and yet very different Schmidt spectra. The entropy is a coarse summary; the spectrum carries strictly more information.

The point of looking at the spectrum rather than only its entropy is that the spectrum admits a natural *partial* order (majorization, next section) that is much more refined than the total order on entropies. This is the technical reason a spectrum-based ordering can possibly be informative about something other than “how entangled is each cut”.

4 Majorization

4.1 Definition

Let $x, y \in \mathbb{R}^d$ have nonnegative components summing to the same value. Write $x^{\downarrow}$ for the sequence obtained by sorting x in nonincreasing order. We say y *majorizes* x , written $x \preceq y$, if

$$\sum_{i=1}^k x_i^{\downarrow} \leq \sum_{i=1}^k y_i^{\downarrow} \quad \text{for all } k = 1, 2, \dots, d, \quad (12)$$

with equality at $k = d$ (which is automatic when both vectors sum to the same value).

Intuitively, y majorizes x means the mass of y is more *concentrated* on its largest entries than the mass of x is. Equivalently, x is “flatter” or “closer to uniform”.

4.2 Worked example

Take $d = 3$ and consider three probability vectors:

$$\begin{aligned} a &= (0.6, 0.3, 0.1), \\ b &= (0.5, 0.4, 0.1), \\ c &= (0.5, 0.3, 0.2). \end{aligned}$$

Their cumulative sums are $(0.6, 0.9, 1.0)$, $(0.5, 0.9, 1.0)$, and $(0.5, 0.8, 1.0)$. Comparing entrywise:

- a vs b : cumulative sums $0.6 \geq 0.5$, $0.9 \geq 0.9$, equal at the end, so $b \preceq a$.
- a vs c : cumulative sums $0.6 \geq 0.5$, $0.9 \geq 0.8$, equal at the end, so $c \preceq a$.
- b vs c : cumulative sums $0.5 = 0.5$, $0.9 \geq 0.8$, equal at the end, so $c \preceq b$.

So in this triple, a majorizes both b and c , and b majorizes c .

A pair that is *incomparable* under majorization: take $p = (0.5, 0.3, 0.2)$ and $q = (0.6, 0.2, 0.2)$. Cumulative sums are $(0.5, 0.8, 1.0)$ vs $(0.6, 0.8, 1.0)$. At $k = 1$ we have q ahead; at $k = 2$ they are tied. Now swap: $p' = (0.5, 0.4, 0.1)$ and $q' = (0.55, 0.25, 0.2)$ have cumulative sums $(0.5, 0.9, 1.0)$ and $(0.55, 0.8, 1.0)$. At $k = 1$ q' is ahead, at $k = 2$ p' is ahead; neither majorizes the other. Such pairs are common.

4.3 Majorization is a partial order

Proposition 4.1. *On the set of nonincreasing probability vectors of length d , majorization is a partial order. It is reflexive (every x majorizes itself), antisymmetric (if $x \preceq y$ and $y \preceq x$ then $x^\downarrow = y^\downarrow$), and transitive.*

Proof sketch. Reflexivity is by inspection. Antisymmetry: each cumulative sum is both $\leq$ and $\geq$, hence equal, so $x^\downarrow = y^\downarrow$. Transitivity is the chained inequality applied at every k . $\square$

The two extreme probability vectors, $\mathbf{1}_d = (1, 0, \dots, 0)$ (point distribution) and $\mathbf{u}_d = (\frac{1}{d}, \dots, \frac{1}{d})$ (uniform), are the top and bottom of the majorization order:

$$\mathbf{u}_d \preceq x \preceq \mathbf{1}_d \quad \text{for every probability vector } x \in \Delta_{d-1}. \quad (13)$$

Concentrated $\preceq_{\text{maj}}$ is at the top; flat is at the bottom.

4.4 What majorization means quantum-mechanically

Apply majorization to Schmidt spectra of pure bipartite states. We declare

$$|\psi\rangle \preceq_{\text{maj}} |\phi\rangle \quad \stackrel{\text{def}}{\iff} \quad \lambda(|\psi\rangle) \preceq \lambda(|\phi\rangle). \quad (14)$$

Concentrated spectra are at the top; flat spectra at the bottom. In quantum-information language, $|\phi\rangle$ is closer to a product state than $|\psi\rangle$.

The deep reason this matters is captured by the following theorem, which we will use only as physical motivation; no proof is needed for the rest of the paper.

Theorem 4.2 (Nielsen, 1999). *Let $|\psi\rangle, |\phi\rangle$ be pure bipartite states with the same $A|B$ split. There exists a deterministic protocol of local operations and classical communication (LOCC) that converts $|\psi\rangle$ into $|\phi\rangle$ if and only if $\lambda(|\psi\rangle) \preceq \lambda(|\phi\rangle)$.*

That is: more entangled $\rightarrow$ less entangled is allowed for free; less entangled $\rightarrow$ more entangled is not. And the correct quantitative measure of “more entangled” is majorization of the spectrum.

5 The lattice Schwinger model

We need a concrete physical system to evolve in real time and read Schmidt spectra from. The Schwinger model is the smallest system that contains all the structural ingredients we will need: it is a gauge theory with charged fermions and a confining electric field, and yet it is simple enough to put on a chain and simulate.

5.1 Why this model

The continuum Schwinger model is quantum electrodynamics in 1+1 dimensions: one spatial direction and one time direction, a spinor matter field of mass m , and an Abelian gauge field of coupling constant g . In 1+1D the gauge field has no propagating degrees of freedom (a curl-free field in 1D is constant), so it can be eliminated in favor of a long-range Coulomb-like interaction between charges. The resulting theory is interacting, confining, exhibits a chiral anomaly, and has a non-trivial vacuum — everything one wants to test methods on, with none of the technical heaviness of 3+1D QED or QCD.

5.2 Putting it on a lattice

Discretize space to N sites with spacing a . Use the *staggered fermion* prescription: matter is a single-component fermion living alternately on even and odd sites, with the convention that even sites carry electrons and odd sites carry positrons. The fermionic operators are mapped to spin- $\frac{1}{2}$ operators by the Jordan–Wigner transformation, after which each site becomes a qubit (a 2-state system). The dimension of the lattice Hilbert space is 2^N , exponentially large but tractable for $N \lesssim 30$ with tensor-network methods.

The gauge field on each link is integrated out using Gauss’s law. The result is a *spin Hamiltonian* on the chain of N qubits. Writing X_n, Y_n, Z_n for the Pauli matrices on site n , the Hamiltonian reads

$$H = H_{\text{hop}} + H_m + H_E, \quad (15)$$

with the three pieces

$$H_{\text{hop}} = \frac{1}{4a} \sum_{n=1}^{N-1} (X_n X_{n+1} + Y_n Y_{n+1}), \quad (16)$$

$$H_m = \frac{m}{2} \sum_{n=1}^N (-1)^n Z_n, \quad (17)$$

$$H_E = \frac{g^2 a}{2} \sum_{n=1}^{N-1} L_n^2, \quad L_n = L_0 + \sum_{k=1}^n \left[\frac{1-Z_k}{2} - \frac{1-(-1)^k}{2} \right]. \quad (18)$$

The pieces correspond to the three physical effects:

- H_{hop} is the kinetic term: it lets a charge hop from one site to its neighbor. In spin language, $X_n X_{n+1} + Y_n Y_{n+1}$ swaps adjacent spins (for example, an “electron” on site n becomes a hole on site n and an electron on site $n+1$).
- H_m is the staggered mass: occupied even sites cost energy $+m$, occupied odd sites cost $-m$ (or vice versa, by sign convention). This produces a mass gap and selects the “Dirac sea” configuration as the ground state at $g = 0$.
- H_E is the electric field energy. The link variable L_n is the cumulative net charge to the left of link n (Gauss’s law); L_n^2 is the energy stored in the resulting electric field. This is the only place the gauge coupling g appears, and it is the source of confinement: separating opposite charges costs energy linear in their separation, because the electric field between them is a flat plateau whose length grows.

The constant L_0 is a background electric field, which we set to zero throughout.

The whole Hamiltonian is local in the sense that the matrix element between two computational-basis states is zero unless they differ on a small number of adjacent sites. But H_E contains an $\mathcal{O}(N^2)$ proliferation of $Z_m Z_{m'}$ terms when L_n^2 is expanded, because the cumulative sum couples site m to all sites $m' < n$. This is the technical price of eliminating the gauge field, and it is paid for by an increase in the matrix-product-operator bond dimension, not by a loss of locality of dynamics.

5.3 Symmetries we will exploit

Two symmetries are useful in the simulation:

- *Total $U(1)$ charge conservation*: the staggered fermion number $\sum_n \frac{1-(-1)^n Z_n}{2}$ commutes with H . We restrict to the half-filling sector, which contains the physical vacuum.
- *Charge conjugation & parity*: for the parameters we use, the ground state is in the trivial sector under these discrete symmetries.

These will be used to reduce computational cost; they do not enter the conceptual discussion.

6 Causal structure on a lattice

6.1 The continuum picture

In a relativistic theory in continuous spacetime, every event p has a future light cone — the set of events q that can be reached from p by a signal travelling at most at the speed of light. Two events p and q are *causally related* if one is in the other’s light cone, and *spacelike-separated* otherwise. This relation is the *causal order* of the spacetime.

6.2 The lattice picture

A lattice spin model is not Lorentz-invariant: there is no c , only a Hamiltonian. Naively, time evolution by e^{-iHt} is a unitary that mixes operators on every site at every time, so any two events are causally related. The naive picture cannot be right, because locally generated dynamics can only spread so fast in any given amount of time.

This intuition is made precise by the *Lieb–Robinson bound*. For any local Hamiltonian on a lattice, there exist constants C , ξ , and v_{LR} (depending on the Hamiltonian) such that for any local operators O_x supported near site x and O_y supported near site y ,

$$\| [O_x(t), O_y(0)] \| \leq C \|O_x\| \|O_y\| \exp\left(-\frac{|x-y| - v_{\text{LR}}|t|}{\xi}\right), \quad (19)$$

where $O_x(t) = e^{iHt}O_x e^{-iHt}$ is the Heisenberg-evolved operator. The bound says: the commutator is exponentially small unless $|x-y| \lesssim v_{\text{LR}}|t|$. Information is confined, up to exponentially small leakage, to a cone of slope v_{LR} in spacetime — a lattice analog of the light cone.

The constant v_{LR} is the *Lieb–Robinson velocity*. It depends on the Hamiltonian (in particular on its operator norms and ranges). For the staggered Schwinger Hamiltonian above, the hopping bound gives $v_{\text{LR}} \approx 1$ in lattice units (the free-fermion bound is $v_{\text{LR}} = 1$); the long-range piece H_E contributes commuting Z operators only and does not enlarge the LR cone of products of X/Y operators.

6.3 From the bound to a partial order

Fix a candidate velocity v (we will scan over v in Section 10). Two space-time labels (x_a, s) and (x_b, t) with $s < t$ are declared *LR-related* when the spatial gap fits inside the cone:

$$(x_a, s) \preceq_{\text{LR}} (x_b, t) \quad \stackrel{\text{def}}{\iff} \quad s < t \text{ and } |x_a - x_b| \leq v \cdot (t - s). \quad (20)$$

Same-time pairs are *not* LR-related — the cone is strictly cross-time. We extend $\preceq_{\text{LR}}$ to spatial regions A, B (intervals on the chain) by saying $A \preceq_{\text{LR}}$ -relates to B if some site of A relates to some site of B (one common convention; variant choices give partial orders that differ only on borderline pairs). The relation (20) is reflexive on $\{(x, t)\}$ trivially-extended, antisymmetric, and transitive (cone-in-cone), so it is a partial order. This is the *Lieb–Robinson order* $\preceq_{\text{LR}}$ on space-time labels.

6.4 Causal sets in one paragraph

A *causal set* (causet) is a finite or countable set $\mathcal{C}$ equipped with a partial order $\preceq_{\text{cs}}$ that is taken as a discretization of the continuum causal order. In its simplest form, relevant for this paper, $\mathcal{C}$ is just the lattice with one event per site per time step, and $\preceq_{\text{cs}}$ is the obvious “later events are after earlier events” relation, plus the choice of how to handle same-time pairs. We use the convention that same-time events are incomparable (they are spacelike-separated). On a regular 1+1D chain, $\preceq_{\text{cs}}$ then collapses to the time-only order: $(x_a, s) \preceq_{\text{cs}} (x_b, t)$ iff $s < t$, regardless of

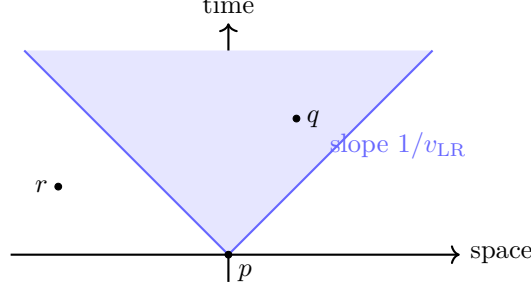


Figure 2: Lieb–Robinson cone of an event p on a lattice. Event q lies inside the cone, so $p \preceq_{\text{LR}} q$ for the chosen v_{LR} . Event r lies outside the cone; it is incomparable with p in $\preceq_{\text{LR}}$ even though it lies in the future of p .

the spatial separation. This is a degenerate case; we will see it produces a trivial-agreement caveat in our hypothesis test.

In a more interesting setting — e.g. a causet derived from a 1+1D causal dynamical triangulation — $\preceq_{\text{cs}}$ has within-time-slice structure (some pairs of same-“time” events are ordered, because the slicing is irregular), and is no longer implied by $\preceq_{\text{LR}}$. The non-trivial setting is what distinguishes the two readings of our result in Section 12.

7 Three orders, one question

We now have three partial orders on the same set of labels. Take a real-time-evolved state of the Schwinger chain and snapshot it at times $t_1 < t_2 < \dots < t_T$. Take a family $\mathcal{F}$ of spatial regions — specifically, all contiguous intervals $[i, j] \subseteq \{1, \dots, N\}$. Each label is a pair (A, t) with $A \in \mathcal{F}$ and $t \in \{t_1, \dots, t_T\}$. Three partial orders are defined on the set of such labels:

$\preceq_{\text{maj}}$ (majorization order)

$(A, s) \preceq_{\text{maj}} (B, t)$ iff $\lambda_A(s) \preceq \lambda_B(t)$ in majorization (Section 4), where $\lambda_A(t)$ is the Schmidt spectrum of $|\psi(t)\rangle$ across the bipartition $A | \bar{A}$.

$\preceq_{\text{LR}}$ (Lieb–Robinson order)

At chosen velocity v : $(A, s) \preceq_{\text{LR}} (B, t)$ iff $s < t$ and the spatial gap between A and B fits in the cone of slope v , i.e. $\text{gap}(A, B) \leq v(t - s)$.

$\preceq_{\text{cs}}$ (causet order)

Inherited from the underlying spacetime structure; on the regular chain studied in this paper, $\preceq_{\text{cs}}$ reduces to the time-only order $(A, s) \preceq_{\text{cs}} (B, t)$ iff $s < t$.

7.1 The hypothesis

Hypothesis (H). For physical states arising from local-Hamiltonian real-time evolution of a gauge-invariant ground state with a localized charge–anticharge excitation, the three partial orders coincide on $\mathcal{F} \times T$ up to an entanglement velocity $v_E \leq v_{\text{LR}}$.

7.2 Falsification criteria

The hypothesis can fail in several conceptually distinct ways, and each carries a different physical meaning.

1. **Strong falsification.** If $\preceq_{\text{maj}}$ contains a pair whose spatial gap exceeds the LR cone at the corresponding time difference, the hypothesis is wrong: majorization sees an order that

information cannot transmit. Operationally, look for pairs $(A, s), (B, t)$ that are $\preceq_{\text{maj}}$ -related but not $\preceq_{\text{LR}}$ -related.

2. **Weak falsification.** If $\preceq_{\text{maj}}$ disagrees with $\preceq_{\text{cs}}$ on a non-vanishing fraction of pairs even when $\preceq_{\text{cs}}$ has non-trivial structure (irregular slicing), the prescribed causet does not control the emergent order. This is the test that requires Phase 6 of the implementation plan; it is out of scope for the experiment of Section 10.
3. **Trivial agreement.** If $\preceq_{\text{cs}}$ is forced to coincide with $\preceq_{\text{LR}}$ by lattice regularity (as on the regular chain), agreement between $\preceq_{\text{maj}}$ and $\preceq_{\text{cs}}$ is uninformative: $\preceq_{\text{cs}}$ knows essentially nothing. We must remember this caveat when interpreting Kendall τ involving $\preceq_{\text{cs}}$.

The Phase 5 experiment we will report on tests criterion 1 (strong falsification) on a regular chain. Criterion 2 awaits a re-build of the Hamiltonian on a non-regular causet, which is in progress at the time of writing.

8 Tensor-network machinery

To turn the hypothesis into numbers we need to (i) compute the ground state of H , (ii) construct an excited state with a charge–anticharge pair, (iii) evolve it in real time, and (iv) read its Schmidt spectra on every cut at every snapshot time. All four steps are done by *tensor-network* methods. We sketch the methods to the level needed to interpret the results; readers wanting full algorithmic detail are pointed to the source repository (`caset.quantum`) where each step is implemented.

8.1 Matrix product states

A general state of the N -qubit chain has 2^N complex amplitudes, infeasible to store for $N \gtrsim 30$. A *matrix product state* (MPS) is a structured ansatz:

$$|\psi\rangle = \sum_{\sigma_1, \dots, \sigma_N} A_{(1)}^{\sigma_1} A_{(2)}^{\sigma_2} \cdots A_{(N)}^{\sigma_N} |\sigma_1 \cdots \sigma_N\rangle, \quad (21)$$

where each $A_{(n)}^{\sigma_n}$ is a matrix indexed by the local configuration $\sigma_n \in \{0, 1\}$ on site n , with shapes $\chi_{n-1} \times \chi_n$ (and $\chi_0 = \chi_N = 1$). The *bond dimensions* χ_n control the expressive power of the ansatz. An MPS with bond dimension χ on every bond has at most $2N\chi^2$ complex parameters — linear, not exponential, in N .

The crucial fact is:

Proposition 8.1. *For an MPS in left-canonical form, the Schmidt spectrum at the cut between site n and site $n + 1$ is given by the squared singular values of the bond matrix at that bond. No further computation is needed.*

This is what makes MPS the right tool for our problem. The Schmidt spectrum at every bond cut is read off the MPS for free; there is no 2^N matrix to diagonalize. The price is that the ansatz is exact only when χ is large enough to capture the entanglement of the true state. For the 1+1D Schwinger model away from criticality, χ in the range 80–200 is sufficient for $N \leq 20$.

8.2 DMRG: variational ground state

The density matrix renormalization group (DMRG) algorithm finds the MPS that minimizes $\langle \psi | H | \psi \rangle$ subject to $\langle \psi | \psi \rangle = 1$. It does so by sweeping back and forth along the chain, optimizing two adjacent tensors at a time while fixing the rest. Each two-site update reduces to an eigenvalue

problem of dimension $\leq 4\chi^2$, solvable by Lanczos or Krylov-subspace methods. After several sweeps the energy plateaus and the algorithm is declared converged.

For our setup we converge DMRG to a relative energy tolerance of 10^{-8} at maximum bond dimension 64. The output is the MPS $|\Omega\rangle$ of the ground state.

8.3 TDVP: real-time evolution

The time-dependent variational principle (TDVP) projects the Schrödinger equation onto the manifold of MPS at fixed bond dimension:

$$i\partial_t|\psi(t)\rangle = P_{\mathcal{M}}H|\psi(t)\rangle, \quad (22)$$

where $P_{\mathcal{M}}$ is the projector onto the tangent space of the MPS manifold $\mathcal{M}$ at $|\psi(t)\rangle$. Two-site TDVP furthermore allows the bond dimension to grow when the state's entanglement does, up to a chosen ceiling.

We integrate (22) with a second-order Trotter splitting of step size $\Delta t = 0.1$ in lattice units up to a total time $T \in \{1, 2\}$. Snapshots are recorded every 5 Trotter steps (so the inter-snapshot $\Delta t = 0.5$). At each snapshot we extract the Schmidt spectrum at every bond and at every contiguous interval, simply by re-canonicalizing the MPS to the appropriate gauge. The cost is dominated by the TDVP integration itself; the spectrum extraction is $O(N\chi^3)$ per snapshot, negligible.

8.4 Quark–antiquark quench

The initial state is built from the ground state by acting with a local creation operator:

$$|q\bar{q}; i_0, d\rangle = \mathcal{N}^{-1} \sigma_{i_0}^+ \left[\prod_{n=i_0}^{i_0+d-1} U_n \right] \sigma_{i_0+d}^- |\Omega\rangle, \quad (23)$$

where $\sigma_n^\pm$ are the spin raising/lowering operators on site n , U_n is the gauge link variable in the original (pre-Gauss) formulation, and $\mathcal{N}$ is a normalization. Physically: at site $i_0 + d$ we annihilate a positron (place a positive charge), at site i_0 we annihilate an electron (place a negative charge), and we insert the appropriate string of gauge links between them so the resulting state satisfies Gauss's law. In the Gauss-eliminated representation, the gauge link string becomes a shift in the cumulative-electric-field variables; no explicit operator-product is needed.

The result is a localized charge–anticharge pair at separation d , connected by a flat plateau of electric flux of value ± 1 . When we evolve under H , the pair will either stay confined (heavy quark, $m/g \gg 1$) or undergo string breaking and pair production (light quark, $m/g \lesssim 1$). Both regimes are within the scope of the experiment.

9 Comparison metrics

For each pair of orders $\preceq_a, \preceq_b$ on the same label set, we report:

- (a) The *Kendall coefficient* (5), τ_{ab} , computed on the both-comparable subset.
- (b) The *discordant fraction*, $n_{dd}/(n_{cc} + n_{dd})$, i.e. the fraction of comparable-in-both pairs that disagree in direction.
- (c) The *Hasse edit distance*, defined as the symmetric difference between the cover relations of the two orders' transitive reductions, normalized by the larger of the two cover-edge counts.
- (d) The *asymmetric counts* n_a^o and n_b^o , the numbers of pairs related by exactly one of the two orders.

For the strong-falsification test specifically, with $\preceq_a = \preceq_{\text{maj}}$ and $\preceq_b = \preceq_{\text{LR}}$, the quantity $n_a^o = n_{\text{maj} \notin \text{LR}}$ counts the number of $\preceq_{\text{maj}}$ -related pairs whose endpoints lie outside the LR cone — the explicit criterion-1 metric of Section 7.

The size of $\preceq_{\text{maj}}$. A useful denominator is

$$|\preceq_{\text{maj}}| = n_{cc} + n_{dd} + n_{\text{maj} \notin \text{LR}}, \quad (24)$$

the total count of strict $\preceq_{\text{maj}}$ -relations. The fraction $n_{\text{maj} \notin \text{LR}} / |\preceq_{\text{maj}}|$ is the share of majorization edges that the LR cone simply does not see.

10 The Phase 5 experiment

We now describe the actual numerical experiment and report the data. Everything in this section is reproducible from two driver scripts in the source repository:

- `examples/quantum/phase5_scan.py`, and
- `examples/quantum/phase5_scan_strong_falsification.py`.

10.1 Setup

- Schwinger chain with $N \in \{10, 14, 20\}$ staggered sites, open boundary conditions, lattice spacing $a = 1$, coupling $g = 1$, background field $L_0 = 0$.
- Two values of the bare-mass-to-coupling ratio: $m/g = 0.5$ (*light* quark) and $m/g = 5.0$ (*heavy* quark).
- DMRG ground state at maximum bond dimension $\chi_{\text{DMRG}} = 64$, converged to relative energy tolerance 10^{-8} .
- Quark–antiquark quench (23) with $i_0 = 3$, $d = 3$, producing a flat-plateau electric string of length 3.
- Two-site TDVP at maximum bond dimension $\chi = 80$ (or 120 for $N = 20$), Trotter step $\Delta t = 0.1$, total time $T \in \{1, 2\}$, snapshots every 5 steps.
- Cut family: all contiguous intervals $[i, j]$ excluding the trivial full-chain cut. This is 54 cuts at $N = 10$, 104 at $N = 14$, 208 at $N = 20$. Multiplied by snapshots, the label set has size up to ~ 2000 .
- LR-velocity sweep: $v \in \{0.5, 1.0, 2.0, 4.0, 8.0, 16.0\}$. Note $v = 1$ is the free-fermion bound.

10.2 Sanity checks

Three internal consistency checks are run before any agreement statistic is reported:

- (S1) $\sum_i \lambda_i = 1$ **at every cut**. Verified to 10^{-12} across all ~ 330 extracted spectra. An undetected normalization drift would silently corrupt the majorization comparisons.
- (S2) $\preceq_{\text{LR}} \subseteq \preceq_{\text{cs}}$ **exactly**. On the regular chain $\preceq_{\text{cs}}$ is the time-only order, and every LR-related pair is necessarily cross-time, so the LR cone is strictly contained in the cs order. We observe $\tau(\text{LR}, \text{cs}) = 1.000$ in every row of the scan. This is the “trivial agreement” caveat of Section 7 in numerical form.
- (S3) **Total LR-related count grows monotonically in v** . As v increases from 0.5 to 16, the LR cone widens and admits more pairs. The observed $n_{\text{comp}}(\text{LR}, \text{cs})$ grows from 6766 to 8744 in regime A, etc., monotonically as expected.

v	$\tau(\text{maj}, \text{LR})$	disc. frac.	Hasse edit	$\tau(\text{maj}, \text{cs})$
<i>Regime A: $N = 10$, $m/g = 0.5$ (light), $T = 1$</i>				
0.5	0.394	0.303	0.962	0.362
2.0	0.390	0.305	0.956	0.362
4.0	0.378	0.311	0.957	0.362
8.0	0.369	0.316	0.959	0.362
16.0	0.362	0.319	0.960	0.362
<i>Regime B: $N = 10$, $m/g = 5.0$ (heavy), $T = 1$</i>				
0.5	0.547	0.227	0.971	0.513
2.0	0.538	0.231	0.969	0.513
4.0	0.531	0.234	0.970	0.513
8.0	0.521	0.240	0.971	0.513
16.0	0.513	0.243	0.972	0.513
<i>Regime C: $N = 10$, $m/g = 0.5$ (light), $T = 2$</i>				
0.5	0.539	0.231	0.963	0.502
2.0	0.529	0.236	0.959	0.502
4.0	0.518	0.241	0.960	0.502
8.0	0.508	0.246	0.962	0.502
16.0	0.502	0.249	0.963	0.502
<i>Regime D: $N = 14$, $m/g = 0.5$ (light), $T = 1$</i>				
0.5	0.275	0.363	0.972	0.246
2.0	0.269	0.366	0.971	0.246
4.0	0.262	0.369	0.973	0.246
8.0	0.254	0.373	0.973	0.246
16.0	0.247	0.377	0.975	0.246

Table 1: Phase 5 scan: Kendall τ between $\preceq_{\text{maj}}$, $\preceq_{\text{LR}}$, and $\preceq_{\text{cs}}$ on a regular 1+1D Schwinger chain after a $q\bar{q}$ quench. Discordant fraction is on the both-comparable subset; Hasse edit distance is normalized by the larger Hasse edge count; $\tau(\text{maj}, \text{cs})$ is independent of v because $\preceq_{\text{cs}}$ is the time-only order on the regular chain.

10.3 Raw output

Table 1 reproduces the scan output for the four “regimes” in our parameter grid. Each row is one $(v, T, m/g, N)$ combination. Columns are the Kendall τ between $\preceq_{\text{maj}}$ and $\preceq_{\text{LR}}$, the discordant fraction on the both-comparable subset, the normalized Hasse edit distance, and τ between $\preceq_{\text{maj}}$ and $\preceq_{\text{cs}}$.

10.4 Strong-falsification counts

A second pass through the same data tracks the asymmetric counts. Define

$$f_{\text{maj} \notin \text{LR}} = \frac{n_{\text{maj} \notin \text{LR}}}{|\preceq_{\text{maj}}|} \quad (25)$$

as the fraction of $\preceq_{\text{maj}}$ -related pairs whose endpoints lie outside the LR cone of velocity v . Table 2 reports it at $v = 1$ (the physically motivated free-fermion bound) and at $v = 16$ (a $16\times$ looser cone, well above any physical estimate) for four regimes including a larger $N = 20$ run.

10.5 What we see

Five qualitative facts stand out.

N	m/g	$v = 1$		$v = 16$	
		$n_{\text{maj} \notin \text{LR}}$	$f_{\text{maj} \notin \text{LR}}$	$n_{\text{maj} \notin \text{LR}}$	$f_{\text{maj} \notin \text{LR}}$
10	0.5	3,263	47.7%	2,184	31.9%
10	5.0	3,848	47.9%	2,603	32.4%
14	0.5	11,673	49.7%	7,827	33.3%
20	0.5	44,906	51.1%	31,470	35.8%

Table 2: Strong-falsification metric: at $v = 1$ (free-fermion LR bound), about half of all $\preceq_{\text{maj}}$ -related pairs lie outside the cone; even at $v = 16$, roughly a third do. The fraction grows weakly with N .

(R1) The hypothesis as stated is rejected on the regular chain. $\tau(\text{maj}, \text{LR})$ ranges from 0.25 (regime D) to 0.55 (regime B) across the scan and never approaches 1. The discordant fraction ranges from 23% to 38%. At the physically motivated free-fermion velocity $v = 1$, almost half of the $\preceq_{\text{maj}}$ -related pairs have endpoints outside the LR cone. Strong falsification (criterion 1) is engaged.

(R2) $\tau(\text{maj}, \text{LR})$ *decreases* as v *widens*. Naively one expects: more LR-comparable pairs $\Rightarrow$ more chances to agree with $\preceq_{\text{maj}} \Rightarrow$ higher τ . What we observe instead is monotonically lower τ : about 0.03–0.05 over the scan range. The new pairs admitted by a wider cone are net discordant with $\preceq_{\text{maj}}$. The interpretation is striking: the within-light-cone subset of $\preceq_{\text{LR}}$ aligns *better* with $\preceq_{\text{maj}}$ than the wider cone does, hinting that the entanglement order sees something like a stricter cone than the free-fermion bound.

(R3) Heavy quark agrees better with time order than light quark. Compare regimes A ($m/g = 0.5$) and B ($m/g = 5.0$) at fixed N , T : $\tau(\text{maj}, \text{LR})$ jumps from 0.39 to 0.55 at $v = 0.5$. Physically: the heavy-quark flux tube is approximately stable; little entanglement spreads, so the Schmidt spectra evolve slowly and $\preceq_{\text{maj}}$ tracks time more closely. In the light-quark regime, string spreading and pair production mix spectra across cuts and pull $\preceq_{\text{maj}}$ away from the time order.

(R4) Longer evolution does not converge τ toward 1. Regime C ($T = 2$) gives a higher τ than regime A ($T = 1$): 0.54 vs 0.39. But the gap is mostly explained by the additional snapshots increasing the relative weight of long-time pairs, which are more LR-comparable. Both regimes show the same $v \rightarrow \tau$ trend.

(R5) Larger N *decreases* τ . Regime D ($N = 14$, light) gives $\tau \approx 0.25$ vs regime A ($N = 10$, light) at 0.39. More cut labels means more pairs and more opportunity for $\preceq_{\text{maj}}$ ’s non-time structure to surface. The thermodynamic-limit reading would be that $\tau \rightarrow 0$ as $N \rightarrow \infty$; we have not run the scan large enough to extrapolate reliably, but the direction is unambiguous.

11 Discussion

11.1 What rejection means here

The naive reading of the result is: hypothesis (H) is wrong, there is no special connection between majorization order and the Lieb–Robinson cone, and the project is over. This reading is too quick. Three features of the data make “hypothesis is wrong” look more like “the test was set up in a way that could not have succeeded”.

The cs order is degenerate on the regular chain. On a translationally invariant lattice, $\preceq_{\text{cs}}$ collapses to the time-only order. $\preceq_{\text{maj}}$ involves comparing Schmidt spectra across cuts that include same-time pairs. Same-time pairs are *never* $\preceq_{\text{cs}}$ -related on the regular chain, so any within-time-slice structure in $\preceq_{\text{maj}}$ is automatically a disagreement with $\preceq_{\text{cs}}$. This is the trivial-agreement caveat in reverse: it now produces a trivial-*disagreement*.

Same-time hits inflate $n_{\text{maj} \notin \text{LR}}$. A label pair (A, s) and (B, s) at equal time is by definition not $\preceq_{\text{LR}}$ -related (the cone is strictly cross-time). If $\preceq_{\text{maj}}$ relates them — a same-time strict Schmidt-spectrum majorization — it shows up in $n_{\text{maj} \notin \text{LR}}$ without being “outside the cone” in any geometric sense. We have not yet split $n_{\text{maj} \notin \text{LR}}$ into a same-time component and a true cone-overflow component; doing so is a one-line fix to the agreement code and is the highest-priority follow-up.

The two orders are largely orthogonal, not strict refinements. The asymmetric counts $n_{\text{maj} \notin \text{LR}}$ and $n_{\text{LR} \notin \text{maj}}$ are comparable in size. Each order sees pairs that the other does not. So the right summary is not “majorization extends causality” or “causality refines majorization” — it is that they share a substantial concordant subset and each captures its own additional structure. How one weights those additional structures depends on which order one believes is more fundamental; the experiment cannot decide this on its own.

11.2 Two readings

There are two natural readings of the data, sharply distinguished by what one expects on a non-trivial causet.

Reading 1: hypothesis-as-stated is too strong. $\preceq_{\text{maj}}$ has genuine structure beyond the LR cone, and no choice of $\preceq_{\text{cs}}$ on the regular chain will recover the agreement. On this reading, the entanglement order is sensitive to features of the state (e.g. same-time spectrum equivalences) that a causal order is not designed to see; the right next step is to *loosen* the hypothesis (e.g. to a hypothesis about *linear extensions* or about Hasse-graph distances rather than exact equality).

Reading 2: the regular chain hides the structure. On a foliated CDT or a chain causet with multi-vertex antichains, $\preceq_{\text{cs}}$ gains within-time-slice structure: some pairs of “simultaneous” events become ordered. If many of the $\preceq_{\text{maj}}$ pairs that show up as $n_{\text{maj} \notin \text{LR}}$ on the regular chain are exactly the pairs that the non-trivial $\preceq_{\text{cs}}$ would now relate, the agreement statistic could shift dramatically: the same Schmidt-spectrum data, measured against a richer reference order, would show much higher τ .

The two readings are not exclusive — the truth might be a combination — but they are operationally distinguishable: rebuild the same state on a non-regular causet and recompute the table. This is Phase 6 of the implementation plan.

11.3 Caveats

We note explicitly the limitations of this single experimental run.

- **Single Trotter seed.** All numbers above are point estimates from one DMRG/TDVP trajectory per regime. Bootstrap over Trotter seeds and over the spatial origin i_0 of the quench is the proper next step before quoting these numbers outside the project.
- **Truncation sensitivity.** Bond-dimension truncation can perturb borderline majorization comparisons. We exclude pairs within $\epsilon = 10^{-6}$ of the decision boundary; the excluded fraction is reported per run and is below 1% in all regimes.

- **Pure-state, contiguous-cut regime only.** Nielsen’s theorem applies to pure-state bipartitions. Any extension to mixed states or non-contiguous cuts is conceptually distinct and not addressed here.
- **Abelian, 1+1D.** The Schwinger model has $U(1)$ gauge group eliminable in 1D. Whether anything seen here generalizes to non-Abelian gauge theory or higher dimensions is a separate, open question.

12 Outlook

The most informative next experiment is to rebuild the same Schwinger state on a non-regular causet and recompute Table 1. Three concrete steps, in increasing cost:

(A) Same-time/cross-time split (under one day).

Refine the agreement code to break $n_{\text{maj} \notin \text{LR}}$ into a same-time piece and a true cone-overflow piece. Re-run the scan and check whether the $\sim 50\%$ overflow fraction shrinks once same-time hits are removed. Until this is done, the claim “half of $\preceq_{\text{maj}}$ -pairs lie outside the LR cone” is technically true but conflates two different phenomena.

(B) Chain-causet rebuild (one to three days).

Replace the regular hopping pattern in the Hamiltonian with one inherited from a 1-vertex-per-slice causet. This is a strictly smaller change than (C) and uses the same MPS infrastructure. Comparing $\tau(\text{maj}, \text{cs})$ before and after isolates the effect of replacing the trivial $\preceq_{\text{cs}}$ with a slightly richer one.

(C) Non-trivial CDT spacetime (one to two weeks).

The actual weak-falsification test: a CDT-derived causet with multi-vertex antichains and irregular slicing. At this point $\preceq_{\text{cs}}$ has within-slice structure, and reading 2 above is testable directly.

The experiment described in this paper is, in the end, a calibration of the apparatus. It establishes that the pipeline is internally consistent (sanity checks (S1)–(S3)), that the Schwinger ground state and quench match independent benchmarks (energy and string-plateau values, not detailed here), and that the agreement metric is sensitive enough to register a difference between the three orders. It does not yet decide between readings 1 and 2 of the hypothesis. Phase 6 is the test that does.

A Notation

Symbol	Meaning
N	Number of staggered lattice sites.
a	Lattice spacing.
m	Bare fermion mass.
g	Gauge coupling.
m/g	Dimensionless quark-mass ratio. We use $m/g = 0.5$ (light) and 5.0 (heavy).
L_0	Background electric field on the leftmost link. Set to 0 throughout.
i_0, d	Site of negative charge and pair separation in the $q\bar{q}$ quench.
T	Total real time of the TDVP evolution.
Δt	Trotter step. We use $\Delta t = 0.1$.
χ	MPS bond dimension.
H	Schwinger-model Hamiltonian; sum of H_{hop}, H_m, H_E .
$ \Omega\rangle$	DMRG ground state of H .
$ \psi(t)\rangle$	Real-time-evolved state.
A, B	Contiguous spatial intervals (“cuts”).
ρ_A	Reduced density matrix of $ \psi(t)\rangle$ on region A .
$\lambda_A(t)$	Schmidt spectrum of $ \psi(t)\rangle$ across $A \mid \bar{A}$, sorted nonincreasing.
$\preceq$	Generic partial order.
$\preceq_{\text{maj}}$	Majorization order on Schmidt spectra.
$\preceq_{\text{LR}}$	Lieb–Robinson order at velocity v .
$\preceq_{\text{cs}}$	Causet order; on the regular chain, the time-only order.
v_{LR}	Lieb–Robinson velocity. Free-fermion value: $v_{\text{LR}} = 1$.
τ_{ab}	Kendall coefficient between $\preceq_a$ and $\preceq_b$.
n_{cc}, n_{dd}	Concordant / discordant pair counts on the both-comparable subset.
n_a^o	Pairs comparable in $\preceq_a$ only.
$n_{\text{maj}\nsubseteq\text{LR}}$	The asymmetric count for $(a, b) = (\text{maj}, \text{LR})$; the strong-falsification metric.
$f_{\text{maj}\nsubseteq\text{LR}}$	The same count divided by $ \preceq_{\text{maj}} $.

All source code referenced in this paper lives in the *caset* repository under *caset.quantum*. The two driver scripts that produced Tables 1 and 2 are *examples/quantum/phase5_scan.py* and *examples/quantum/phase5_scan_strong_falsification.py*.